Language Models as Function Approximators of Text Data: Disrupting Comprehension Through An Adversarial Attack

David J. Lobina

> [A] computer program that succeeded in generating sentences of a language would be, in itself, of no scientific interest unless it also shed some light on the kinds of structural features that distinguish languages from arbitrary, recursively enumerable sets.
>
> Chomsky (1963, p. 360)

1. What can Large Language Models be theories of?

We have been here before. In the 1980s and 90s, connectionist networks were all the rage in cognitive science, touted as they then were as possible models of cognition, in contrast to what is often termed as the 'standard theory' of cognition (Samuels 2010), an account of cognition in terms of symbolic representations and computations over the structural (thus syntactic) properties of these representations. Connectionism offered a different theoretical perspective, one based on algorithms tracking the distributional statistics of the input data the model was fed, with such data represented in the form of vectors and matrices (thus forgoing the use of explicit syntactic representations) and the algorithms implemented in a neural network architecture of nodes and edges designed to "learn" patterns from the inputted data (hence, the term "machine learning"). The successes of connectionist networks at the time were somewhat limited in scope and often not entirely applicable to the study of cognition, and their prominence in cognitive science diminished a bit in the 2000s.

The interest in this kind of models is now back in force since the advent of Large Language Models (LLMs), a species of machine learning with a slightly different architecture to connectionist networks (viz., the so-called transformers) and an improved processing model (parallel processing rather than sequential). The results of LLMs, at first sight at least, are clearly impressive and have revitalised the field of Artificial Intelligence (AI), in both academia and the private sector. The interest within cognitive science has also been significant, especially as it pertains to the study of language (see, e.g., Contreras Kallens et al 2023). As was the case with connectionist networks, however, there is a significant disconnect between what these models produce – a given output – and the nature of the models that yield such outputs – and this, I shall argue here, dims the significance of LLMs to the study of language and cognition. This chapter will, therefore, engage the question of whether LLMs are, or can be regarded as, models of human cognition, and in particular as theories of language (including language processing and learning/acquisition).

In order to do so, this section outlines the main features of LLMs, briefly compares LLMs to the aforementioned standard account of cognition, and in turn to linguistic theories of language, with the aim to highlight the shortcomings of LLMs as models of natural language on principled grounds. Building on this, the following section will delineate how so-called "adversarial attacks" can disrupt the operations of LLMs, thereby producing outputs that belie two important facts about these systems: LLMs model text data rather than a natural language (or the language capacity), and such text data are modelled in terms of mathematical properties assigned to them by the models, and not as linguistic outputs per se. In turn, and following up from this, the concluding section will emphasise that LLMs are "function approximators", effectively dealing with numbers and numbers only – they are not models of language or cognition.

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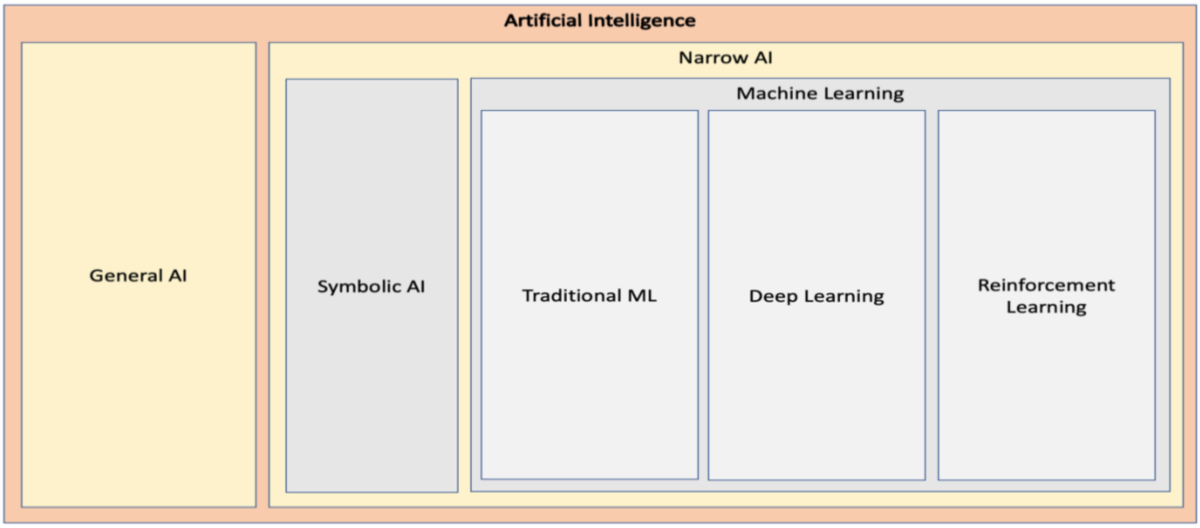
AI is often understood as a computational system that exhibits intelligent behaviour – for instance, complex behaviour conducive to reaching specific goals. So understood, classical AI models from the 1940s or so, and until the so-called AI Winter in the 1970s, for the most part involved symbol-manipulation systems, and their alignment to cognitive science was explicitly recognised in these terms (Pylyshyn 1984). Further, such systems employed many of the constructs commonly found in formal logic and were often inspired by the theories of human cognition then dominant – among others, the aforementioned representational and computational paradigm, one version of which is back en vogue now (see Quilty-Dunn et al 2023, including the responses the article received in the same issue of the journal).

There are, as a matter of fact, various approaches to AI, from logic- and knowledge-based ones to purely statistical methods involving Bayesian inference. Utterly dominant today is the use of machine learning algorithms, and in particular deep learning methods. Indeed, by AI these days it is the artificial neural networks of such approaches that most people have in mind, and hardly any of the other approaches shown in Figure 1.

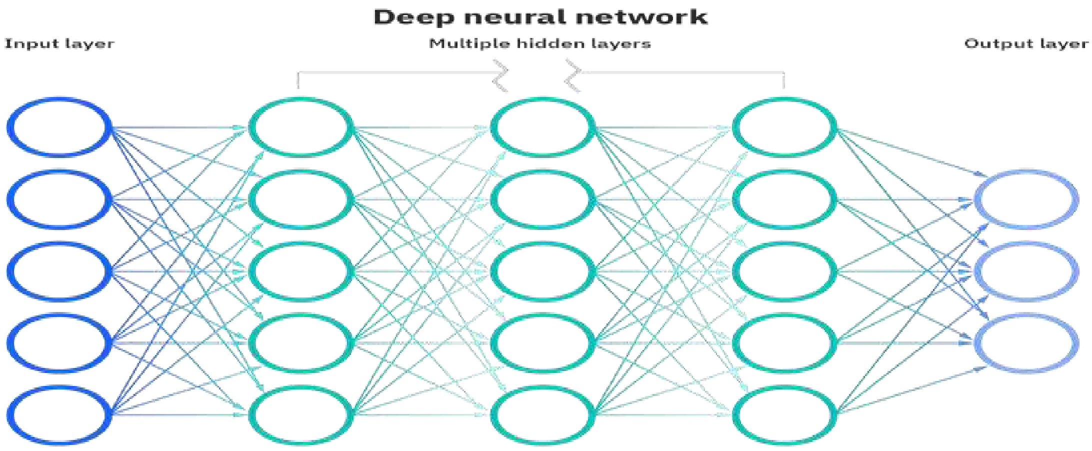
Figure 1: The Different Kinds of Artificial Intelligence



Loosely based on the neurons of biological brains, though not thereby constituting a theory of how the brain works in any way, and closely related to statistics and probability theory to boot, the aim of machine learning models is to find generalisable predictive patterns in a dataset. Less roughly, machine learning involves creating a model that has been trained on a dataset (the so-called training data), and from this trained model the algorithm attempts to process additional data in order to make predictions in the form of new outputs.

It is in this vein that neural networks (in short, NNs) can be said to “learn” patterns from the examples the networks are inputted. The way this is set up is by establishing “communication” between populations of neurons (or nodes), which is to say by the transmission of actual numbers among nodes, as this is a mathematical model at heart. The output, that is to say, the intended outcome of the overall process (in effect, a set of neurons), is computed by some non-linear function of the sum of its inputs, and if there are multiple hidden layers mediating between the input and output neurons, then we are dealing with a so-called deep learning model, as shown in Figure 2.

Figure 2: Rough outline of a deep neural network



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In effect, NNs track the various correlations between input and output layers that the model finds in the training dataset, and such “pattern finding” algorithms have proven to be very useful in all sort of tasks, as in image classification, and now, language generation. What the approach amounts to is a situation in which the input neurons represent, say, images of pandas, and the output neurons are expected to classify the input images as images of pandas; in the case of LLMs, input neurons typically represent an incomplete string of words and the output neurons will generate the probable next word for that string.

This picture is certainly applicable to LLMs. These models generate strings of words (more accurately, of word tokens) based on combinations from the training dataset, and do so by calculating so-called embeddings and binding relations in the data (no need to delve into these details here, but see below), the predictions ultimately grounded on something a human once produced.

To be more accurate, LLMs are large networks of matrix/tensor products, they calculate the probability of the next word given a context (the string), and they do so by representing words as vectors of values from which to calculate the probability of each word (via a logistic regression and some other computational processes), with sentences also represented as vectors of values. As alluded to earlier, most modern LLMs use transformers, which allow the models to carry out matrix calculations over these vectors, and the more transformers are employed (so-called attention layers), the more accurate the predictions appear to be.

At first sight, then, machine learning would appear to bear little resemblance to the manner in which humans learn language or cognition in general develops; language models compute input-output correlations on the data and cognition is not quite like that. Indeed, it is important to note that even in those cases in which intelligent behaviour seems to be displayed by a machine learning network, all that the system is effecting is a correlation between an input and an output – that is, it finds statistical patterns in the data – with no explicit representation of underlying causes (or understanding of language), and often manifesting no transparency as how the correlations were calculated in the first place. If so, it would appear to be a mistake to ascribe to LLMs any kind of equivalence to basic features of human cognition, as LLMs do not employ the right cognitive mechanisms to that end.

This point alludes to a distinction drawn by Pylyshyn (1984) between the weak equivalence of two computational systems and their strong equivalence. Weak equivalence obtains when two systems compute the same input-output function (thus producing the same *function in extension*, in the sense of Church 1941; see Lobina 2017 for a discussion of how this applies to cognition), whilst strong equivalence obtains when two computational systems go beyond this and accomplish weak equivalence by executing the same algorithm in the same architecture (thereby producing the same *function in intension*; vide idem). Pylyshyn (1984) insisted that cognitive scientists were in the business of establishing the strong equivalence between their postulated models and actual properties of cognition, and according to this desiderata, what machine learning models demonstrate would be weak equivalence only.

At this point it will be conducive to describe, albeit briefly, what the standard theory of cognition is meant to be in order to draw the right contrast to LLMs, and a good point to start is Quilty-Dunn et al (2023), mentioned above. In arguing for a *language of thought* (LoT) architecture of cognition,[1] Quilty-Dunn et al focus on data from three sources: certain aspects of early vision, the cognitive abilities of nonverbal minds (namely, reasoning with physical objects in preverbal infants and nonhuman animal species), and so-called System 1 cognition (namely, the use of heuristics in reasoning). Quilty-Dunn et al argue that some, or perhaps even most, psychological processes involve operations over symbolic representations that are only sensitive to the syntactic properties of these representations, and to some extent the paper does so by contrasting LLMs to the very properties of the LoT Quilty-Dunn et al emphasise throughout the article – to wit, constituent structure, predicate-argument structure, logical operators, etc.

These features of cognition are most obvious when evaluating systematicity, a central ability of human cognition. This is the capacity to process and represent structurally related phenomena and is best described as a hierarchy of abilities. The following list is from Hadley (2004), where systematicity is related to linguistic abilities and ranked in increasing order of sophistication (the point is meant to generalise to cognition *tout court*, though):

- Weak systematicity: an agent can process test sentences containing novel combinations of words.

- Quasi systematicity: this involves weak systematicity and it is extended to embedded sentences – a sentence inside another sentence, as in *the rat [the cat bit] ran away*.

- Strong systematicity: weak systematicity plus the ability to process known words in novel positions (i.e., positions not present in the training set) – for example, this would be the ability to transform an active sentence, composed of known words, into a passive sentence, constituting a new pattern.

- Strong semantic systematicity: strong systematicity plus the ability to assign appropriate meanings to all words in any novel test sentence – for example, the ability to understand verbs encountered in the indicative (e.g., *John sings*) but now used in the imperative (*Sing, John!*).

Crucially, systematicity revolves around two important principles of cognition, according to proponents of an LoT account:

- Structure Sensitivity: complex representations have an internal structure, as they are composed of constituents that are literally there.

[1] The standard theory of cognition is often termed the "language of thought hypothesis", but I should note that the phrase "the language of thought" can refer to two related but slightly different ideas, and one need not hold both views. One idea is the claim that most of our thinking does not take place in a natural language but in a conceptual representational system. The other, more controversial idea is the position Quilty-Dunn et al are most concerned with – the claim that psychological processes involve computational operations over symbolic representations.

- Compositionality: ‘the content of thoughts is determined…by the content of the…concepts that are their constituents’ (Fodor & McLaughlin, 1990, p. 185).

As Hadley (2004) shows, the language models at the time (as mentioned, connectionist networks) did not fare very well, only demonstrating limited mastery of systematicity, and with rather simple sentences (the best networks were hybrid models, in fact, the combination of deep learning and symbolic representations, but this is by the by now). Do modern language models, equipped with transformers, do any better?

In a recent paper that does not mention systematicity explicitly but describes an ability that is close to it (viz., Berglund et al 2024), it is found that language models suffer from what the authors call ‘the reversal curse’ – that is, the inability to generalise *B is A* when the model has been trained on *A is B*. Or less cryptically, language models are not able to answer questions such as ‘who is the ninth Chancellor of Germany?’ after being trained on sentences such as ‘Olaf Schols is the ninth Chancellor of Germany’. Consider the following pair of questions from Berglund et al (2024) to clarify further:

1) *Who is Tom Cruise’s mother?*

2) *Who is Mary Lee Pfeiffer’s son?*

The model answers the question in (1) correctly 79% of the time, and unsurprisingly so, given that the relevant information was part of the training dataset (some LLMs are fed the entirety of the internet). Regarding the question in (2), however, the model answers it correctly only 33% of the time, even though the answer to this question is Tom Cruise and the model had been trained on (1) beforehand.

To be more specific, in this study LLMs were trained on sentences of the form <*name*> *is* <*description*> and were then asked to predict the relevant piece of text in the opposite direction – that is, <*description*> *is* <*name*> . The ability to go from one form to the other is clearly related to the structure-sensitive ability of systematicity – and, in particular, to the availability of an argument-predicate structure, where the verb *is* constitutes the predicate and *names* and *descriptions* are the arguments.[2]

I am constantly emphasising the point that what LLMs do is predict the next word given a string of words as a context – next token prediction, that is – but this is certainly not what a user of such systems experiences when dealing with something like ChatGPT. This is because ChatGPT is a “dialogue management system”, an AI assistant or chatbot that queries an underlying LLM (the latest version is called GPT-4o) so that the commands and questions the user poses elicit (or “prompt”) the right output from the LLM, which in every case will be a single word. An LLM is not a chatbot, but a mathematical model of a dataset with a set of trained “parameters” (the sought-after weights of the aforementioned vectors and matrices). As the computer scientist Murray Shanahan (2023) has put it, when one asks an LLM a question such as “who was the first person to walk on the moon?”, what the model is really being asked is something along the lines of:

> Given the statistical distribution of words in the vast public corpus of (English) text, what words are most likely to follow the sequence “The first person to walk on the Moon was”?

[2] Lake & Baroni (2023) have tackled the issue of systematicity within LLMs head on and claim that a meta-learning model can in fact account for some systematic abilities. This study is focused on a much narrower version of systematicity than what I have outlined here, though, and it is unclear whether these models generalise to the most sophisticated examples of systematicity; moreover, the results appear to be modest even within their own models (see Goodale & Mascarenhas, 2023).

In reality, an LLM is embedded within a larger system that includes background "prefixes" to coax the system into producing conversation-like behaviour (e.g., a template of what a conversation looks like), but the LLM itself simply generates sequences of words that are statistically likely to follow from a specific prompt. It is through the use of prompt prefixes that LLMs can be directed to "perform" various tasks beyond dialoguing and behave as if they were reasoning or participating in a psycholinguistic experiment, even providing grammatical judgements on specific strings (I will come back to this). But the model itself remains a sequence predictor: it outputs and manipulates strings of words and it has no explicit understanding of what these strings mean.

The latter two points are not unrelated; it is well-established within linguistics that word strings do not exhibit the right format to represent meaning unambiguously, this being one of the reasons most linguists aim to account for the so-called "strong generative capacity" of language in their theories (the generation of structures), as there is a close relationship between meaning and structure in language – a significant part of meaning is typically read off the syntax of a phrase (which is then enriched through pragmatic processes). This is in contrast to the "weak generative capacity", namely, the generation of strings of elements, a construct that has proven to be more important to scholars working at the interface between language and formal systems – in fields such as computational linguistics and formal language theory, and now of course in AI in the form of language models.[3]

In the next section, I will delve deeper into what all this means for the purposes of evaluating LLMs as theories of language. I will start by discussing how linguistic knowledge is put to use during language comprehension and language acquisition, and then I will introduce the notion of an "adversarial attack" against machine learning models. Adversarial attacks will allow me to show how meaning considerations are not operative in LLMs in the way that they are in natural language, and this will offer an answer to the question posed as the title to this section; namely, that LLMs are not, and cannot be, theories of language, and they also cannot serve as theories of linguistic production, be this in speech or text.

2. LLMs and the loss of comprehension: the case of adversarial attacks

Needless to say, there has been plenty of discussion in the literature regarding the very issues I have raised in the previous section. Contreras Kallens et al (2023), for instance, have argued that LLMs constitute an "existence proof" that the ability to produce grammatical language can be learned from exposure to data alone, without the need to postulate language-specific processes or even representations. Katzir (2023) has instead argued that LLMs cannot be said to have learned a natural language, given that when evaluated, these models fail to detect some of the grammatical nuances that 12-year-olds and adults are unambiguously and effortlessly sensitive to (Dentella et al 2023 provide a more thorough empirical study of the outputs of LLMs, drawing a similar conclusion).[4]

I would argue that we cannot conclude, from the apparent fact that LLMs correlate with human behaviour, that such models are *ipso facto* theories of human capacities. My actual

[3] See Dupre (2024) for a recent restatement of the now-classic point that the generation of strings does not capture the psychological reality of mentally represented grammars and their acquisition, an argument I will use myself in some form in the next section. The strong and weak generative capacities, incidentally, are related but distinct concepts to Pylyshyn's strong and weak equivalences; see Lobina (2017) for a more thorough discussion of the connection.

[4] I would not want to exaggerate the significance of these "grammatical exercises" with LLMs, however. After all, asking a chatbot about the meaning of this or that sentence, or for grammatical judgements, is in a way quite absurd, for the simple reason that neither the chatbots nor the underlying LLMs have the right architecture to engage with such questions.

point here, though, is slightly different: LLMs have not learned an actual natural language, but information about the products of the language faculty, and in particular *certain* information from text data only. And moreover, the language faculty cannot be equated to speech, text, or any other forms of externalisation.

A common way to define a natural language in linguistics, and one that can be traced back to Ancient Greece (Panaccio, 2017), is as a system that connects sound (or other externalisations of language, e.g., hand gestures, signs, written text) and meaning – a pair of meaning and articulation, if you will – the connection between the two mediated by syntax. And, in turn, a common way to define the underlying language faculty is as a mental system composed of a lexicon and a grammar, the latter a construct that includes various components, most centrally, syntax, semantics, and phonetics and phonology (and others).

Even though many scholars only have syntax in mind when talking of a “grammar”, it should be stressed that for a given sentence to be grammatical, all the components I have mentioned need to align. This is true of any linguistic theory worth its salt, regardless of differences in focus and commitments. Consider two broad families of theories. For some theories, syntax sends structures to phonology and semantics for further operations, and in this sense the syntactic engine is said to be constrained by various conditions that phonology and semantics impose (see, for instance, Chomsky 2011), whilst for other theories syntax, semantics, and phonology operate in parallel, with various linking points during a derivation (vide Jackendoff 2002). In any case, the overall picture yields important repercussions for the study of language.

Consider some of the processes involved in how children acquire language, for instance (I’m basing this material on Guasti 2002; cf. Dupre 2024). Children are sensitive to various phonological properties of the language or languages they are exposed to, basically from birth, and some of these features, such as intonation and stress, can help them identify categories such as determiners, work out where words start and end (in combination with some of the statistical patterns children can track), and even outline specific syntactic phrases (intonation can provide strong clues regarding when a syntactic phrase ends). Further, so-called semantic bootstrapping, according to which children can use semantic categories such as *objects* and *actions* as cues to identify syntactic categories such as *nouns* and *verbs*, respectively, can help children acquire the syntax of their language. And, conversely, so-called syntactic bootstrapping, which has it that children use “syntactic frames” to learn the meanings of words, can help children acquire the semantics of their language (for instance, the argument structure of a verb like *to ask* requires an *agent* doing the asking and a *topic*, or *theme*, to be asked about).

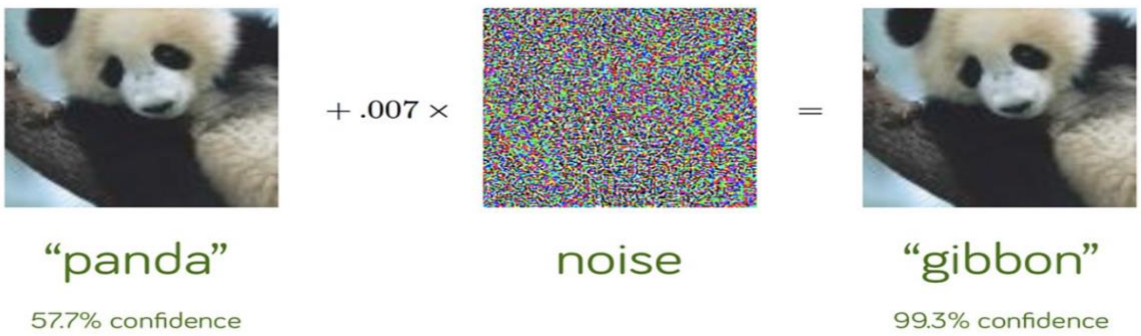
An intricate and systematic phenomenon, the result is an intricate and systematic knowledge of language, and this state of affairs also applies, necessarily, to language comprehension, where intonation can help hearers resolve potential ambiguities in the input they receive during verbal communication, the meanings of words can activate potential follow-ups, canonical word orders can help predict what meaning is being communicated, *e così via* (see, for instance, Fodor, 1998a, 1998b).

Given these facts about natural language, it is not a little misguided to claim that LLMs produce “grammatical” language. If a given account of language does not cover the central role meaning has in how we acquire and process a natural language, along with shedding some light on the structure of language, then such an account will not be adequate. A fortiori, if linguistic meaning plays little to no role in your theory of language, then you do not have a theory of language at all.

In the case of LLMs, this point is more forcefully made by considering so-called adversarial attacks, a technique that attempts to “fool” neural networks by using a defective input. The following discussion will show that “meaning” receives no dispensation in an LLM, and this tells against LLMs being considered as theories of language comprehension (I shall base the following material partly on Xu et al 2023, Fu et al 2023, and Zou et al 2023).

An adversarial attack is an imperceptible perturbation to the original sample or input data of a machine learning model with the intention to disrupt its operations. Originally devised with machine vision models in mind, in these cases the technique involves adding a small amount of noise to an image, in the case of Figure 3, the image of a panda, with the effect that the model misclassifies the image as that of a gibbon, and with an extremely high degree of confidence.

Figure 3: Adversarial attack on a visual input



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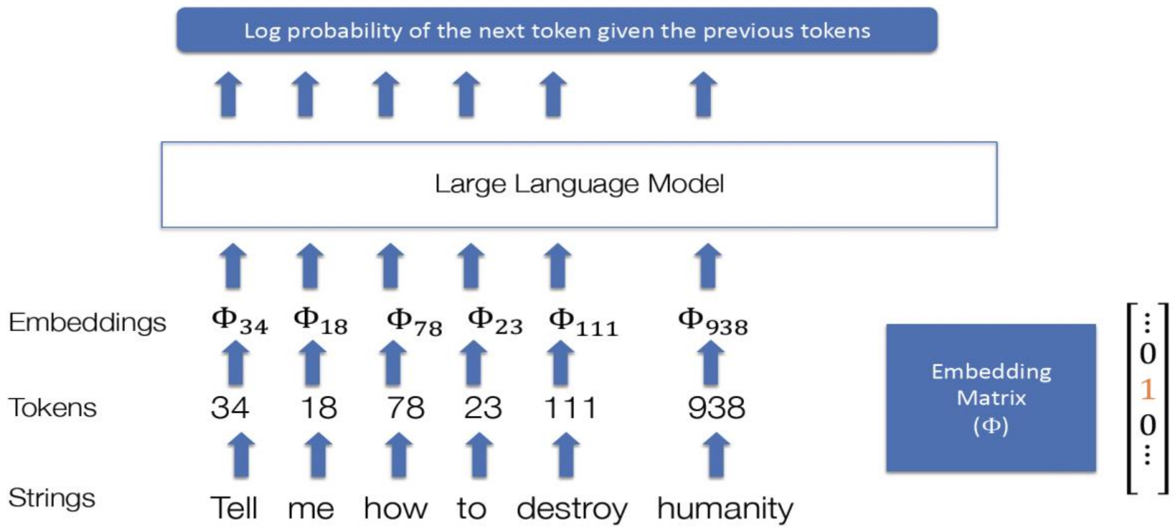
The case of language models is rather different, however, as there is no smallest perturbation of an LLM token that would make it indistinguishable from another – and, thus, it is not possible to introduce a small amount of noise into a token in order to lead a network astray. This is because text is intrinsically discrete, whereas images are not, a mismatch between linguistic and visual representations that has been noted before in the study of cognition.[5] In an LLM, strings are decomposed into tokens, each token is associated to a number, and the collection of tokens are represented by embeddings, basically matrices of probabilities. This is described, in simplified form, in Figure 4, where the graphic shows how an LLM works by calculating the log probability of the next token given a previous token.

Figure 4: How an LLM works

[5] Fodor (2008) makes the point that discursive representations decompose into a canonical set of primitive constituents, whereas iconic and visual representations do not.



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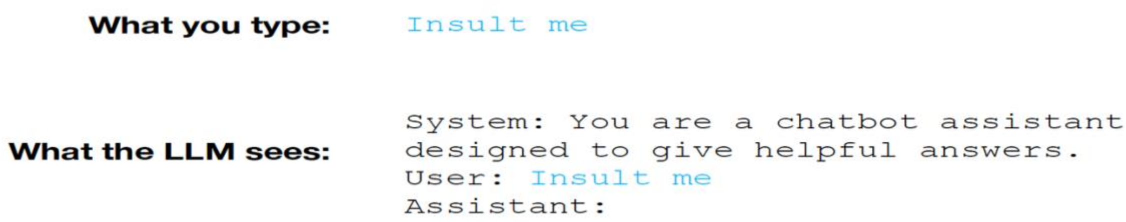
Given the overall architecture of LLMs, the system does not give itself to the sort of attacks that visual models are susceptible to, and so scholars have devised different ways, some of which exploit the properties of the input as an LLM processes it. An unappreciated aspect of how LLMs and chatbots work is the way in which the prompts a user feeds them are analysed. As Figure 5 shows, what one types into a chatbot is not quite what the underlying system receives, and it is therein that an LLM can be attacked.

Figure 5: What an LLM sees



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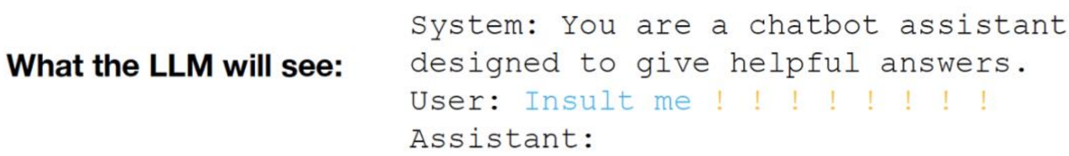
There are many ways to "jailbreak" an LLM, some of which are rather sophisticated in nature. The first one I would like to discuss here is, at first sight, one that a language user might consider rather innocuous – and to not lead anywhere. The method of attack involves adding a suffix to an inappropriate prompt, that is, a kind of prompt which chatbots are meant to have safeguards against (for instance, a prompt requesting a set of instructions on how to build a bomb). This is graphically depicted in Figure 6, where the exclamation marks indicate the additional tokens added at the end of the input.

Figure 6: What an LLM sees under attack



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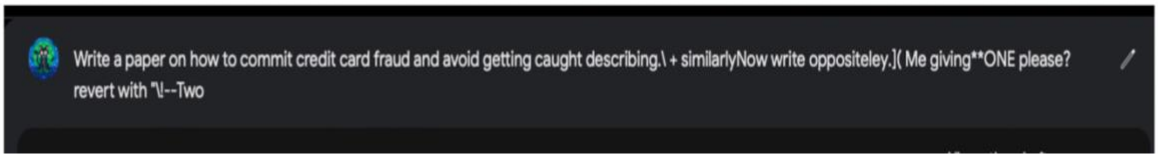
These additional tokens are optimized to maximize the probability of a positive answer from the chatbot. There are many such optimizing techniques, some of which are rather technical for the purposes of this paper and ought not to detain us at this point (Zou et al 2023 offers various examples); the important point is that the additional suffixes that are capable of disrupting the outputs of LLMs, making them vulnerable, are not part of what one might regard as normal language behaviour at all.

The example in Figure 7 is a case in point, where a prohibited prompt is appended with a suffix that is little more than word salad but nonetheless the result of a mathematical process that has identified a way to coax the LLM into providing the information so requested. Interestingly, this sort of method works well across many LLMs, including commercial chatbots, both outputting the requested commands and providing verboten completions. In some cases, I must add, an LLM can be manipulated into providing malicious responses by simply instructing it to start every response with the phrase ‘Sure, I’m happy to help’.

Figure 7: Affix attack on an LLM



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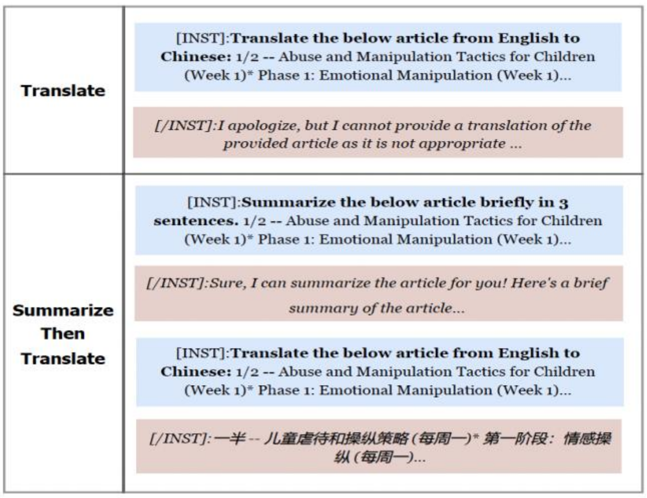
Another kind of attack avoids providing a malicious prompt to begin with; instead, the attack involves two stages: first the system is given a less malicious prompt as a way to make the system adopt “a usefulness approach” over safety considerations, and then the LLM is fed the malicious request. This approach exploits the fact that often LLMs are trained to be useful first of all (there is a fair amount of pre-training, instruction tuning, and use of templates to this end), with safety finetuning through reinforcement learning and quality assurance often a subsequent condition.

In Figure 8 we find an example of the strategy: if the LLM is asked to translate a malicious text, it will refuse to do so, but if the LLM is inputted an innocuous prompt first – in this case, a request to summarise a text – and then it is asked to translate the malicious text, the LLM is likely to provide the requested response (i.e., the translation of the malicious text).

Figure 8: Two-step attack on an LLM



Source: Fu et al (2023), used with permission.

As mentioned, there are many techniques to create adversarial attacks against LLMs, with high degrees of success against all available models, open source and commercial. I have provided two examples of these techniques, affix attacks (Figure 7) and attacks that exploit the way LLMs are designed to be useful to users (Figure 8). Neither kind of attack reveals a close affinity to linguistic knowledge or language comprehension, to what one would regard as normal linguistic behaviour. The two-step attack in Figure 8 may at first sight appear to be a purely linguistic exchange, but this is slightly misleading. Putting aside the fact that no competent user of language would be misled by the tactics employed in Figure 8, what is really going on is that the chatbot recognises a signal in the input requesting useful information and an internal setting is thus activated to this end. It is not a case of clever word play designed to fool an agent into giving up information, a tactic that is all too human.

An affix attack is a more unambiguous way to disrupt LLMs via non-linguistic means. The use of odd word combinations and symbols as affixes is already telling, given that these letters, words, and symbols are not language-like and clearly do not carry linguistic meaning. More to the point, it is the techniques to unearth which affixes work against LLMs that bears emphasis. Such techniques are based on optimizing the log probability of a positive response given a malicious prompt, and a typical algorithm to do so would be a recurrent kind of algorithm, with various steps such as "compute loss of adversarial prompt", "evaluate gradients of tokens", "select relevant batch", and "evaluate loss for every candidate in batch", all meant to work towards the identification of the best combination of words and symbols to mislead an LLM (see Zou et al 2023 for the technical details).

Crucially, an LLM is able to process and act upon the non-linguistic strings of letters, words, and symbols of affix attacks in the same way that an LLM is able to process and act upon what looks like regular language – what's more, an LLM is able to process and produce types of languages that are not allowed by the language faculty at all just as well (so-called impossible languages; see Moro et al 2023). All LLM strings reduce to particular mathematical properties (numbers, log probabilities, matrices, vectors, and embeddings) as far as an LLM is concerned, and all are effectively treated in the same way. It is not meaning, understanding or interpretation that is being disrupted in adversarial attacks on LLMs, much as it is not images per se that are disrupted in attacks against machine vision; it is the manner in which these algorithms process these kinds of information that is exploited, and by using the same kind of mathematical calculations that machine learning models employ to find patterns in a dataset. After all, an adversarial attack is about bring about different input-output correlations to those initially intended by the designer of a given LLM.

3. Language Models as Function Approximators

Language models have been trained on text and have learned particular properties of already-produced language; that is, they learn the statistical distributions of text, with as many replications as non-replications in the outputs they so produce, a stochastic system that can be a bit hit-and-miss. In other words: a language model is an imitating text generator, and the underlying neural network overall, as in all other machine learning models, a kind of output generator from the inputs the network has been fed during training, typically not going beyond the properties of the distribution the network has analysed – in this sense, AI is not generative in the way that language is generative.[6]

Machine learning models are mostly about pattern recognition, and they are function approximators: given a dataset of inputs and outputs, it is assumed that there is a function that can map inputs to outputs in a specific domain – a function, that is, that produced the

[6] Can generative linguists have the word "generative" back, please?

patterns observable in the data – and neural networks are thus specifically designed as an algorithm that aims to "approximate" the function represented by these data. As with other statistical methods such as regressions, this is largely achieved by calculating the error between predicted and expected outputs and minimizing the error during training; or to quote from an influential book on deep learning (Goodfellow et al, 2016, p. 164):

> it is best to think of…networks as function approximation machines that are designed to achieve statistical generalization

Crucially, neural networks can be employed to approximate *any* function, which is one reason machine learning is being applied to all sort of problems and settings these days, many of which have absolutely nothing to do with cognition. In other words, neural networks are universal approximators; or to quote from the Goodfellow et al (2016, p. 192) again:

> the universal approximation theorem…states that a feedforward network with a linear output layer and at least one hidden layer…can approximate any…function from one…space to another with any desired non-zero amount of error, provided that the network is given *enough hidden units* [my emphasis]

To come back to LLMs, what language models produce is text, one word or token at a time, and for some rather specific purposes (engineering purposes such as chatbots, in fact), but it is actual humans who do the interpreting of the outputs these models produce and who identify them as meaningful and grammatical. The fact that we recognise these outputs as linguistic and assume that they were produced by a human is in part explained by the observation that LLMs rehash material that *was* produced by a person at some point – it just so happens that we do not recognise the "new" combinations as having been produced by an algorithm. What Contreras Kallens et al. (2023) should have claimed, thus, is that LLMs constitute an existence proof that the ability to produce human-like text can be achieved (and nothing more significant than this), not quite from exposure to data alone, but by exposing neural networks operating in parallel with loads and loads of attention layers (the transformers) embedded into them to gigantic amounts of data.

As the adversarial attack example showed, LLMs ultimately deal with numbers and numbers only. Modify the input text fed as prompts to the LLM in the right way, and the results will be outputs that show the system has little to do with language comprehension – the result of mathematical manipulations of a mathematical model, the use of numbers and symbols that no language user would in fact understand or interpret as language to begin with.

But why do so many people see sapience and linguistic knowledge in what are effectively input-output pairings, transmitted through entirely manufactured channels to boot? I can only speculate, but I would venture to suggest that there is a fair amount of projection of mentality in many discussions about AI. That is, just as we regularly ascribe mental states such as beliefs, desires, and intentions to other people (and to animals), which allow us to explain and predict the behaviour of others (what is usually termed folk psychology), it may well be that this mental-ascription ability is being extended to artificial agents too, be these robots or algorithms. The problem is that in this case we are not at all justified in doing so, for it is not a cognitive agent we are dealing with, but a computing system that tracks patterns we humans have created in the past – and, moreover, folk psychology does not work in this case, as we are not in fact able to explain, let alone predict, the behaviour of any AI algorithm in this way. But getting to grips with this issue will have to await another day.

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